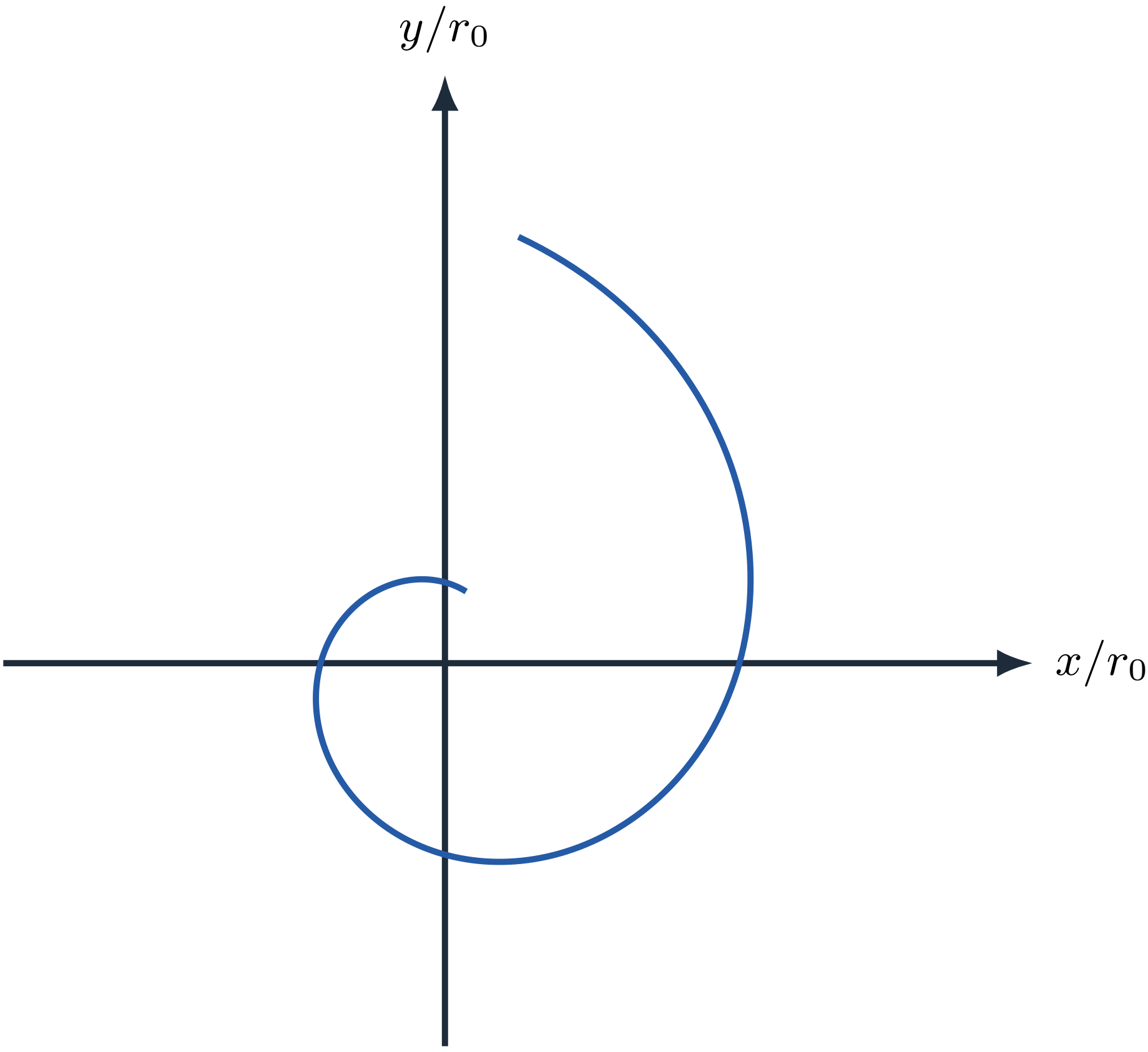
y/r_0
x/r_0